

Exp No: 4

7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS

OBJECTIVES

- To understand the interfacing and working of a 7-segment display with the AT89C51 microcontroller.
- To write and execute an assembly language program to display digits on the 7-segment display.
- To simulate and verify the 7-segment display operation using Proteus 8.

PROGRAM

ORG 000H

UP:MOV P2,#0C0H

ACALL DELAY

MOV P2,#0F9H

ACALL DELAY

MOV P2,#0A4H

ACALL DELAY

MOV P2,#0B0H

ACALL DELAY

MOV P2,#99H

ACALL DELAY

MOV P2,#92H

ACALL DELAY

MOV P2,#82H

ACALL DELAY

MOV P2,#0F8H

ACALL DELAY

MOV P2, #80H

ACALL DELAY

MOV P2,#90H

ACALL DELAY

DELAY: MOV R5,#10

H1:MOV R4,#180

H2:MOV R3,#255

H3:DJNZ R3,H3

DJNZ R4,H2

DJNZ R5,H1

RET

END

OUTPUT

